

GHIBRAN ESA ANUGRAH

Telecommunications Engineering | Politeknik Negeri Semarang |
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PROFESSIONAL SUMMARY

Telecommunications Engineering graduate from Politeknik Negeri Semarang with hands-on experience in fiber optic infrastructure, FTTx networks, RF engineering, computer networking, embedded systems, IoT, and digital signal processing. Developed practical engineering experience through internships at PT Telkom Akses, technical projects, professional training, and organizational leadership. Proficient in applying telecommunications and engineering principles to system development, technical troubleshooting, and network infrastructure. Strong in analytical problem solving, project coordination, and technical documentation, with career interests in telecommunications, network engineering, RF systems, IoT, and embedded technology.

EDUCATION

POLITEKNIK NEGERI SEMARANG

Diploma III in Telecommunications Engineering
Tembalang, Semarang, Indonesia

September 2023 – September 2026

SMK NEGERI 1 SLAWI

Computer and Network Engineering
Slawi, Tegal, Indonesia

July 2020 – April 2023

WORK EXPERIENCE

PT TELKOM AKSES

Intern Staff | September 2025 – December 2025
Tegal, Indonesia

- Performed maintenance and repair of network infrastructure, including OLT, ODC, ODP, and fiber optic systems, to support network stability and service quality.
- Assisted with FTTx installation, configuration, and troubleshooting in accordance with Telkom Akses technical standards.
- Prepared technical documentation and field reports to support operational evaluation and coordination with supervisors.

PT TELKOM AKSES

Intern Staff | June 2022 – September 2022
Slawi, Indonesia

- Assisted field technicians with the installation of new FTTx services, including IndiHome and IndiBiz, for residential and business customers.
- Supported customer network maintenance and repair activities, including equipment replacement and signal quality inspection.
- Participated in onsite troubleshooting to maintain the quality of internet, telephone, and IPTV services.

PROJECT EXPERIENCE

IoT-Based Smart Camera Cabinet with Automated Hysteresis Climate Control

Internet of Things (IoT) & Embedded Systems | 2026

- Developed an IoT-based camera storage cabinet using ESP32, SHT31, Peltier cooling, and automated humidity control.
- Implemented hysteresis-based humidity regulation, real-time web monitoring, REST API communication, and scheduled UVC sterilization.
- Achieved 99.21% humidity control success in a non-AC room and 97.22% in an AC room, with stable operation for more than 144 hours.

Water Resource Monitoring & Billing Management System

Web Application / Smart Water Management System | 2025

- Developed a web-based water management system for digital meter recording, automated billing, and real-time monitoring.
- Implemented water usage analytics, customer management, historical records, and leak/anomaly monitoring using PHP and MySQL.

Radar Target Detection Simulator via Matched Filtering

Digital Signal Processing / Radar Systems | 2025

- Developed a radar signal processing simulator using LFM chirp generation, multi-target echo simulation, AWGN noise modeling, and matched filtering.
- Implemented peak detection and target range estimation, successfully detecting three simulated targets with zero range estimation error.

OFDM Wireless Communication Simulator

Wireless Communications / Digital Signal Processing | 2025

- Developed an end-to-end OFDM communication simulator using QPSK, 16-QAM, and 64-QAM modulation with AWGN and Rayleigh channel models.
- Implemented FFT/IFFT processing, Zero-Forcing equalization, and BER analysis to evaluate wireless transmission performance under different SNR conditions.

ECG Heart Rate Detection & Arrhythmia Analysis System

Biomedical Signal Processing / Digital Signal Processing | 2026

- Developed an ECG signal processing system using GNU Octave to analyze recordings from the MIT-BIH Arrhythmia Database.
- Implemented bandpass filtering, adaptive R-peak detection, RR interval analysis, heart rate estimation, and basic bradycardia/tachycardia classification.

Microstrip Antenna Design Projects

RF Engineering / Antenna Design | 2025

- Designed and simulated single, 2×2, and 4×1 microstrip antenna configurations for the 902–928 MHz ISM band using CST Studio Suite.
- Evaluated S11, VSWR, gain, directivity, and radiation patterns to assess antenna performance for sub-GHz wireless communication applications.
- Achieved S11 values around –28 dB with VSWR of approximately 1.1–1.3 across the developed antenna configurations.

CERTIFICATIONS

Radio Frequency Engineer - 193107501110-44/VSGA/BLSDM.Komdigi/2025

Vocational School Graduate Academy (VSGA), Digital Talent Scholarship 2025

Balai Pelatihan Sumber Daya Manusia Komunikasi dan Digital (BLSDM Komdigi) Yogyakarta

CCNA: Switching, Routing and Wireless Essentials

Cisco Networking Academy

CCNA: Introduction to Networks

Cisco Networking Academy

KKNI Level II – Computer and Network Engineering Competency - 61100 3513 2 0000258 2023

Badan Nasional Sertifikasi Profesi (BNSP)

Offshore Oil and Gas Processing – Free Class - FreeClass/1905/2024/311

Engineering Academy by Anak Teknik Indonesia

ORGANIZATIONAL EXPERIENCE

KELUARGA MAHASISWA BIDIKMISI POLITEKNIK NEGERI SEMARANG

President | September 2024 – September 2025

- Established organizational strategies and coordinated programs from planning and implementation to evaluation.
- Managed academic and non-academic development programs to support member development in leadership and entrepreneurship.
- Represented the organization in establishing relationships and collaborations with internal and external stakeholders.

HIMPUNAN MAHASISWA ISLAM KOMISARIAT POLINES

General Secretary | June 2025 – Present

- Managed organizational administration and systematic record-keeping, including official correspondence, meeting minutes, and formal documentation.
- Coordinated administrative communication among executive members, organizational members, and external stakeholders.
- Prepared meeting agendas and periodic accountability reports to support transparent and organized operations.

AWARDS & ACHIEVEMENTS

- 1st Place – Sultan Agung Literacy Fest Essay Competition 2025, Badan Eksekutif Mahasiswa Universitas Sultan Agung
- 2nd Place – Information Network Cabling, Tegal Regency Student Competency Competition 2022, Musyawarah Kerja Kepala Sekolah SMK Kabupaten Tegal
- Silver Medal & Bronze Medal – Pekan Essay Nasional (PENA) 2, Universitas Dhyana Pura
- Best Paper – National Essay Competition 2025, Universitas Negeri Semarang Scholarship Community
- 2nd Runner-Up – Fair Essay Competition 2025, Forum Komunikasi Mahasiswa Politeknik Indonesia, Politeknik Negeri Semarang
- Finalist – Idea Festival 5 2026, Sentosa Foundation and Institut Pertanian Bogor
- Finalist – Faperta Fair 9 2026, Sentosa Foundation and Universitas Muhammadiyah Mataram
- Finalist – National Essay Competition (NESCO) 2 2026, PRISMA and participating universities

TECHNICAL SKILLS

Telecommunications & Networking	: RF & Cellular Networks, Fiber Optic & FTTx, Computer Networking, Routing & Switching, Wireless Networking, Network Troubleshooting
Embedded Systems & IoT	: ESP32, Arduino, C/C++, IoT Systems, Sensors & Actuators, HTTP/API Communication, MQTT, System Integration
RF & Signal Processing	: RF Engineering, Digital Signal Processing, OFDM, Radar Signal Processing, Matched Filtering, Microstrip Antenna Design
Software & Tools	: Cisco, MikroTik/ The Dude, Cacti, PRTG, Wireshark, Nmap, Atoll, Pathloss, TEMS Investigation, OptiSystem, GNU Octave, Proteus, CST Studio Suite
Web & Database	: PHP, JavaScript, HTML/CSS, MySQL

PROFESSIONAL SKILLS

Problem Solving & Technical Troubleshooting · Leadership & Team Coordination · Project Management · Analytical Thinking · Communication & Stakeholder Coordination · Technical Documentation

LANGUAGES

Indonesian	: Native
English	: Basic